



DVN36104 SPR 2026

Data Visualisation and Narratives
Spring 2026
Gerard Kelly





36104 · DATA VISUALISATION AND NARRATIVES

Seeing Data

Visualisation, intelligence, and interpretation

How do charts create meaning?

Choose privately before groups form

For the later group assignment, choose whether you prefer:

- a group where approved AI-assisted recording or transcription may be used;
- a group where those tools will not be used; or
- no preference.

No response carries an academic advantage or penalty.
Responses are visible to authorised teaching staff, not to other students, and are used only to support group allocation.



Canvas → Get started

A compatible-group preference is not blanket consent to recording.

The authorised workflow has boundaries

The subject outline and each assessment brief define what is permitted. For notebook activities, the proposed workflow is GitHub Copilot in VS Code, subject to written authorisation and these safeguards:

- do not submit personal, confidential, restricted or unpublished assessment data;
- treat generated code and explanations as unverified drafts;
- retain prompts or prompt summaries and disclose significant use;
- test outputs, explain accepted code and remain responsible for every submitted claim;
- follow the stricter rule wherever an assessment is marked AI-restricted.

Educational authorisation is not the same as institutional procurement or security approval. Confirm current UTS and faculty requirements in writing.



THE PREMISE

**A chart is a claim about
how the world is organised.**

Five decisions produce the claim

- **Selection:** which data, which window, which denominator.
- **Encoding:** which visual channels carry which variables.
- **Comparison:** what is placed next to what.
- **Framing:** title, caption, annotation, baseline.
- **Omission:** everything left out, silently.

Opening question for every artefact this semester:

what has the designer chosen to make visible, and what remains invisible?

LEARNING OUTCOMES

By the end of today

1. Explain how selection, encoding and framing turn data into meaning.
2. Separate observation from interpretation, in your own reading and in generated text.
3. Explain why identical summary statistics can conceal different structures.
4. Use an AI assistant to produce a chart, then verify every claim it makes.

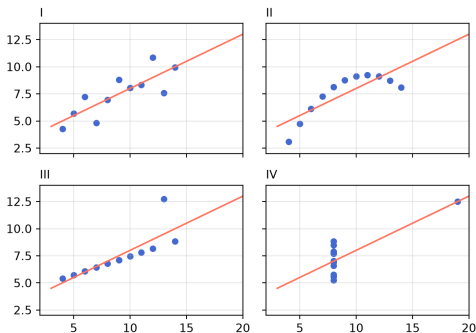
Four datasets, one set of statistics

	I	II	III	IV
mean of x	9.0	9.0	9.0	9.0
mean of y	7.50	7.50	7.50	7.50
variance of x	11.0	11.0	11.0	11.0
correlation	0.816	0.816	0.816	0.816
regression line	$y = 3.00 + 0.50x$			

Prediction: will the four datasets look essentially the same when plotted?
Write yes or no, and why, before the next slide.

Looking changes the conclusion

Anscombe's quartet — identical statistics, different shapes



Course reproduction of Anscombe (1973). A summary statistic is not a substitute for a picture.

The chain we will use all semester

data → **encoding** → **observation** → **insight** → **narrative**

- **data**: records of the world, already shaped by collection;
- **encoding**: the mapping of values onto marks and channels;
- **observation**: what any careful reader can see (“the chart shows”);
- **insight**: an interpretation that survives verification (“this suggests, because”);
- **narrative**: insights arranged to support a decision.

Every failure we study this semester is a broken link in this chain.



THE CLAIM SPRINT

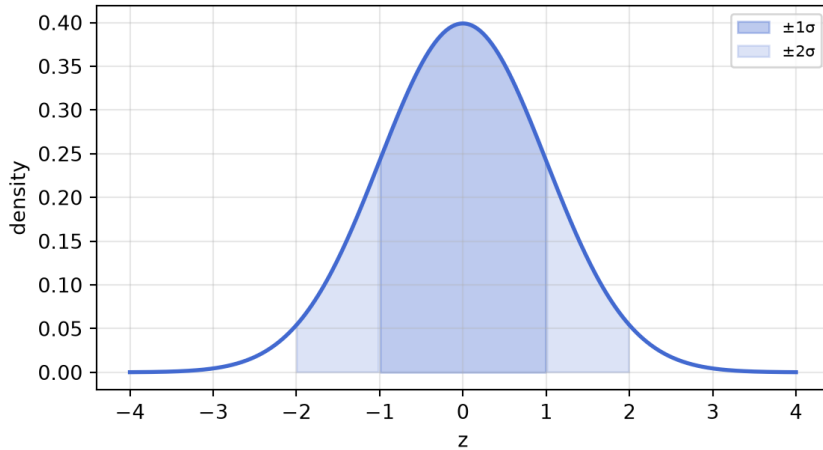
Twelve charts - 40 seconds each.

For any two: write one “the chart shows ...” sentence.

All figures: course reproductions, generated from data. We debrief two of them.

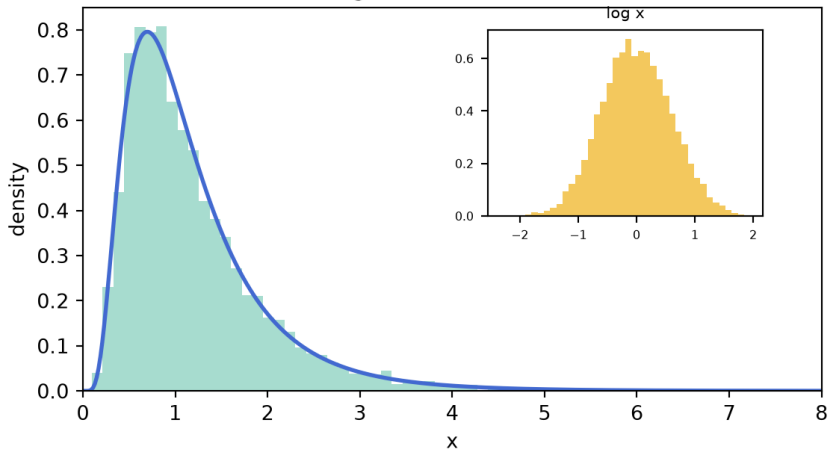
The bell curve

Standard normal distribution



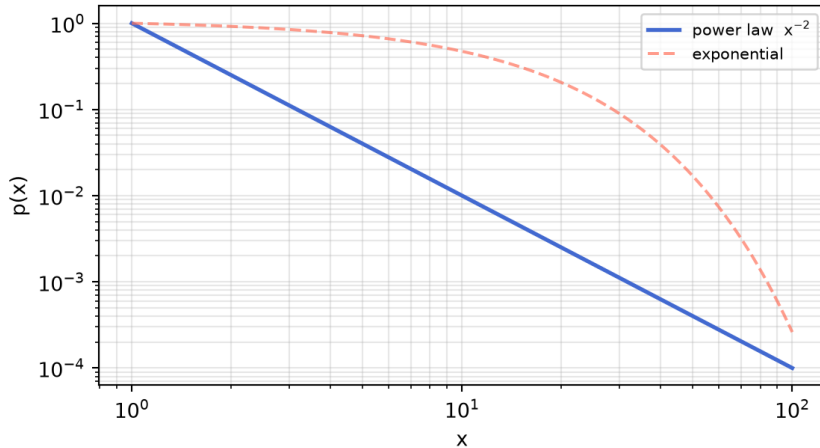
The log-normal

Log-normal distribution

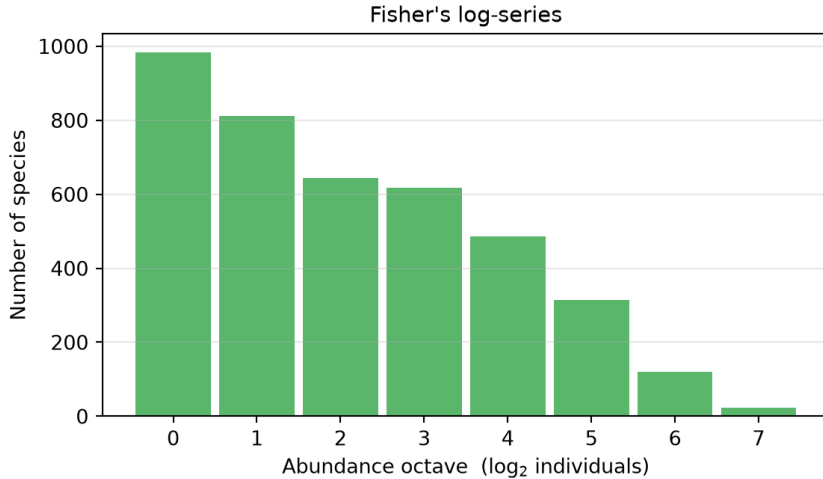


The power law

Heavy tail on log-log axes

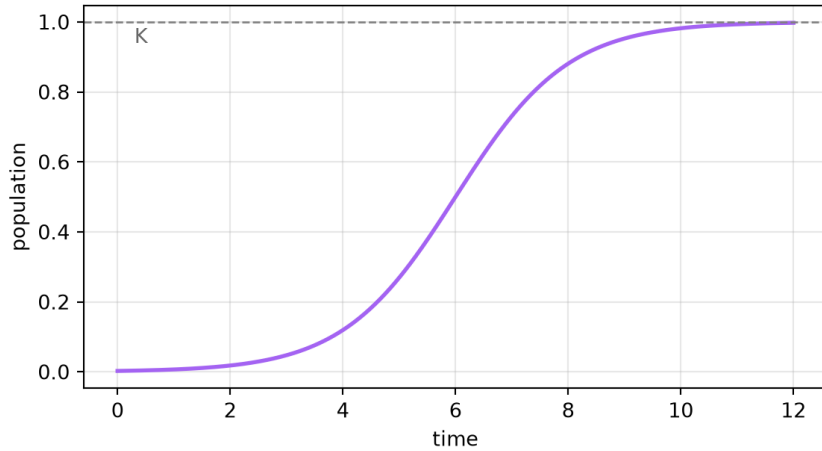


Species abundance

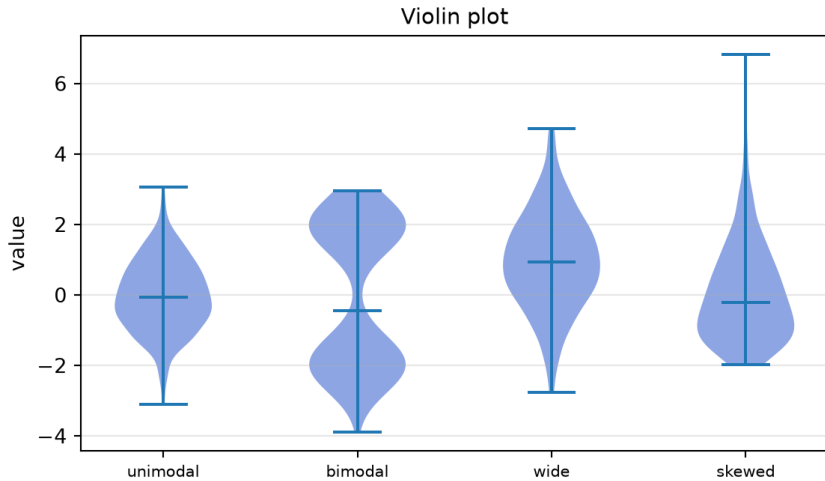


The S-curve

Logistic (Verhulst) growth

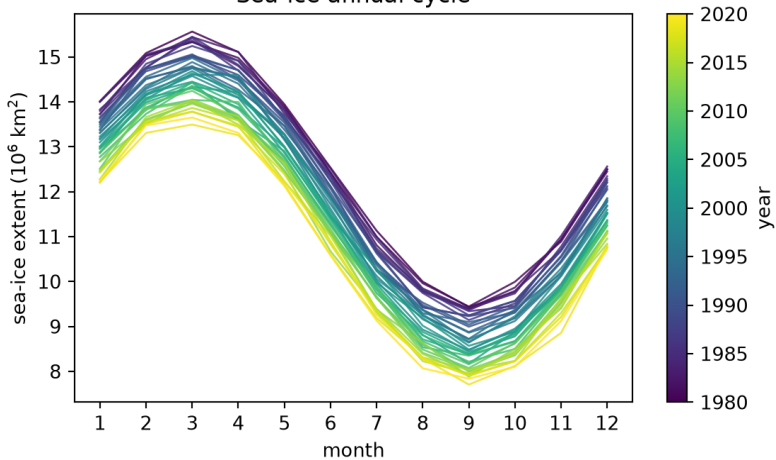


The violin plot



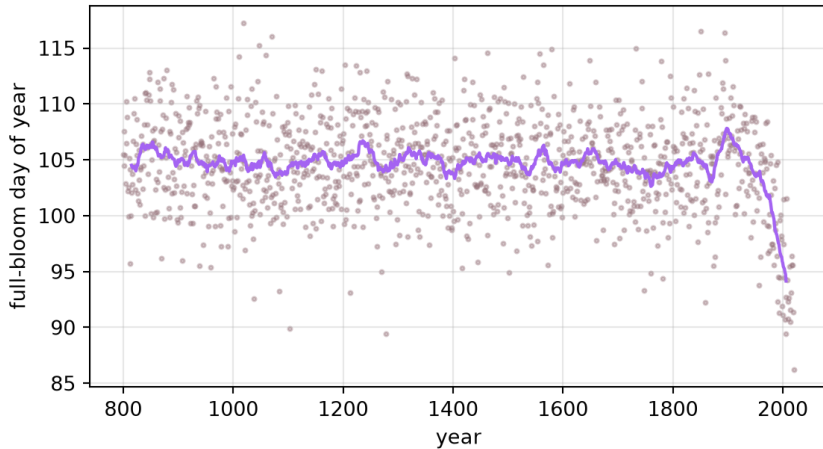
Arctic sea ice

Sea-ice annual cycle

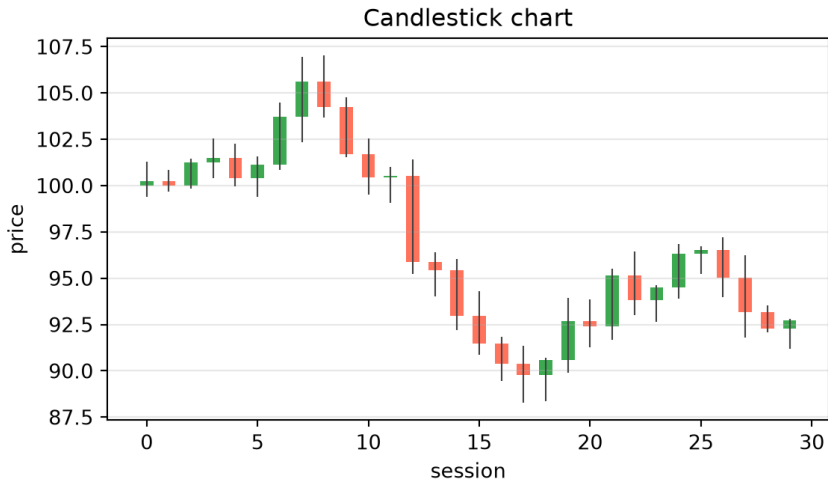


Kyoto cherry blossom

Kyoto cherry-blossom dates

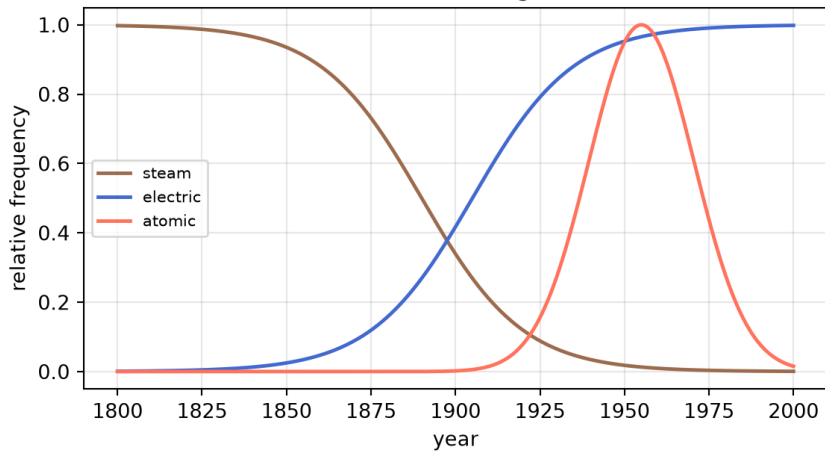


The candlestick chart



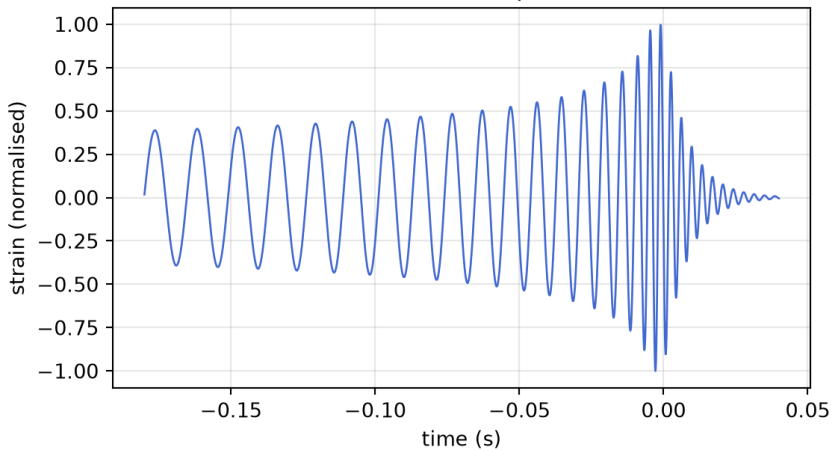
Google Books Ngram

Culturomics / Ngram

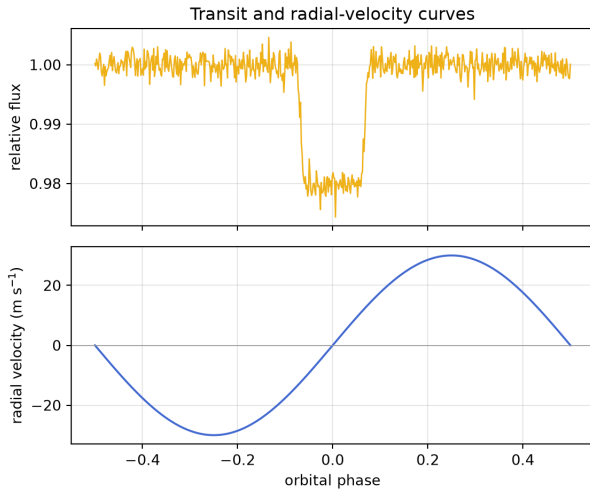


A gravitational wave

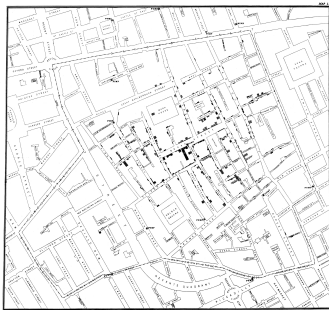
Gravitational-wave chirp (GW150914)



An exoplanet transit



A picture can become an argument



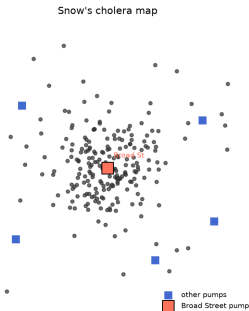
John Snow, cholera deaths in Soho (1854); published 1855. Public domain.

Each black bar: one death,
stacked at an address.

Read it with the chain:

- what is the **encoding**?
- what is the strongest **observation**?
- what **insight** is Snow selling?

Pattern, inference, and the missing evidence



Course reproduction: deaths cluster on the Broad Street pump.

- **Pattern:** deaths cluster around one pump. Visible.
- **Inference:** the water is the vehicle. Not visible: argued.
- **Causal evidence** came from elsewhere: the brewery workers who drank beer; the workhouse with its own well; deaths that stopped after the handle came off.

The map did not prove the theory.
It made the theory **impossible to ignore**.

Two kinds of sentence

- **“The chart shows ...”**, reserved for what is directly visible:
deaths stack along Broad Street; the cluster thins with distance from the pump.
- **“This suggests ...”**: interpretation:
the pump water carries the disease; the map vindicates removing the handle.

Most of what a chart “says” is supplied by the reader.

This class is about noticing when that happens, in you, and in a machine.

Six ways a true chart misleads

Every number checked, every label true, and the impression still wrong.

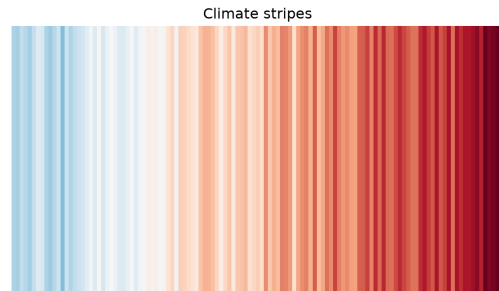
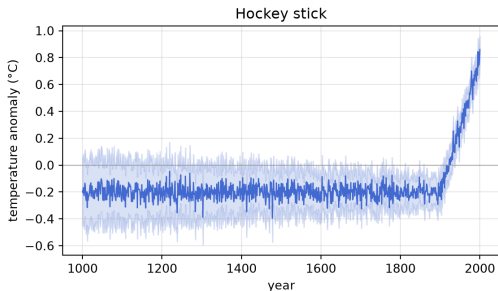
- truncated or missing baseline
- cherry-picked time window
- dual axes tuned to a story
- area standing in for value
- binning and smoothing choices
- omitted denominators

After Cairo, *How Charts Lie* (required reading, selected chapter).

HOW CHARTS LIE

The window is part of the claim

Crop either one to 1998–2012 and you can sell a pause. The axis window is an editorial decision.

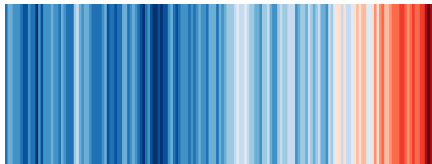


The hockey stick: context makes the rise the story.

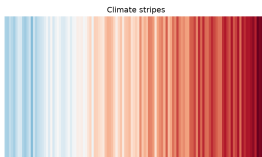
Climate stripes: the same claim, at a glance.

WARMING STRIPES

The original, and the redraw



Ed Hawkins, warming stripes, global 1850–2018 (Climate Lab Book). CC BY-SA 4.0.



Course reproduction, generated from data, same data, same claim, reproducible.



VISUALISATION + AI

**Fluent interpretation is not
valid interpretation.**

What the assistant does well, and doesn't

- An AI system can describe a pattern **confidently** without checking whether the data supports the claim.
- Generated text is optimised to read well. Reading well and being right are different properties.
- The failure mode is not usually a wrong number, it is an **unsupported claim** delivered in a supported tone.

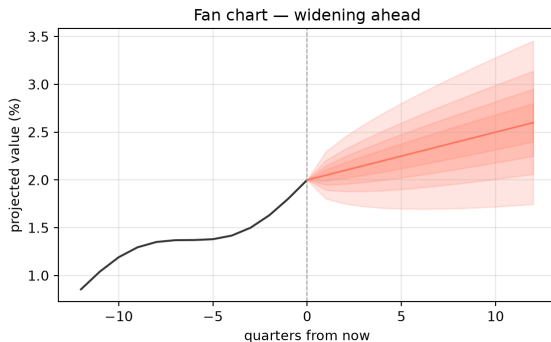
Your job is not to avoid the assistant. It is to **own the verification**.

The verification ladder

1. **Source:** what data was actually used? Can you point to it?
2. **Transformation:** what was filtered, aggregated, reshaped? Check counts and types.
3. **Visibility:** is each claimed pattern actually visible in the plotted data?
4. **Support:** classify each claim: **supported / plausible but unverified / unsupported / contradicted**.
5. **Disclosure:** record what the tool contributed and what remains unverified.

This ladder is the spine of the notebook today, and of Assessment 1, Part C.

Uncertainty drawn rather than suppressed



A fan chart admits what it does not know. Generated interpretations rarely volunteer this, you must ask.

5 MINUTES SOLO · 5 IN PAIRS · 5 AS A ROOM

Think · pair · share

Find a chart published **this week**, news, social, a report.

- What claim does it make?
- Which of the six mechanisms should worry you?
- What would you need to verify before repeating its claim?

Keep the chart. It may become your Assessment 1 artefact.



BREAK · 10 MINUTES

**When we return:
the notebook lab, then the studio.**

Notebook lab · 45 minutes

1. **Anscombe**: summary statistics vs the picture.
2. **Generate and verify**, assistant-drawn chart of the transport dataset, then the checks: rows, types, missing values, totals.
3. **Claim audit**: a generated interpretation, every sentence classified: supported / plausible but unverified / unsupported / contradicted.

The contract: **predict** before you generate · give the assistant a **bounded task** · **verify** before you accept · **disclose** at the end.

Blocked on syntax? Ask for the starter notebook, that is what it is for.

Studio · Critique and Repair · 25 minutes

- **Draw three chart cards** (pack in activities/): for each, the claim, the audience, and one way it could mislead.
- **Read** the satisfaction artefact: main claim, audience, what is directly visible.
- **Test** the design: encodings, easy comparisons, what is aggregated or hidden.
- **Verify and repair**: what you would check, an alternative view, a repaired caption.
- **Responsibility**: who benefits, and who carries the cost if it is wrong?

Ten minutes solo, ten with a partner, five to revise.

Pair exchange · the Critique 1 frame

Swap artefacts: one **human-made chart**, one **AI-generated interpretation**.

- Same questions for both: claim, audience, task, comparison, omission.
- Then the AI-specific one: **which claims would survive the ladder?**

This is exactly Assessment 1, Parts A and C, at rehearsal scale:

Part A critique (supervised, in studio) · Part B redesign ×3 (next class) · Part C claim audit.

Close the loop

Look back at your Critique and Repair sheet.

Which of your sentences were observations? Which were interpretations?

- One design decision I made today:
- One risk or limitation I noticed:
- One thing I need to verify before next class:

Diagnostic quiz on Canvas after class, open book, closed assistant, ungraded. It calibrates the course, not your mark. Next class: Choosing Visual

Forms: what visual task are we performing?

Image credits and reading

- Snow (1855) image: public domain, via Wikimedia Commons.
- Warming stripes: Ed Hawkins, Climate Lab Book, CC BY-SA 4.0.
- All other figures: course reproductions, generated from data.

Required reading: Cairo, *How Charts Lie* (ch.); D'Ignazio & Klein, *Data Feminism* (ch.).